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EXPERIENCE

- SochDB** Remote
Core Contributor – Embedded AI-Native Database April 2026 - July 2026
 - Core Engine Contributions:** Core contributor to SochDB, an embedded Rust database unifying HNSW vector search, MVCC, and ACID transactions as a single storage engine for AI agent memory, replacing brittle vector-DB + relational-DB stacks with one embeddable primitive.
 - State-of-the-Art Retrieval Benchmarking:** Owned end-to-end LoCoMo retrieval benchmarking, pushing scores to 97.17% Hit@200 / 94.84% Recall@200 (84.77%/79.57% @10, 91.50%/86.88% @20) via a candidate-retrieval + cross-encoder reranking pipeline, exceeding published results from competing memory-layer startups.
 - Benchmark Infrastructure & Competitive Positioning:** Built the benchmark harness and competitive-analysis tooling behind SochDB's retrieval claims, directly informing founder-level pitch and fundraising materials on retrieval quality vs. competitors.
- Stealth AI Startup** Remote
Early Engineering Contributor – AI Agents & Evaluation May 2025 - August 2025
 - LLM Workflow Prototyping:** Built 4 LLM workflow prototypes for early-stage AI product experiments, integrating LLM APIs, prompt orchestration, backend API calls, and application logic for internal MVP validation.
 - LLM Evaluation:** Developed DeepEval-based evaluation scripts covering 120 test cases across 18 prompt and workflow variants, measuring hallucination, answer relevance, and task-completion quality during rapid iteration.
 - Deployment Support:** Containerized and deployed 2 Next.js application prototypes using Docker and Azure Container Registry, reducing repeat demo setup from 7 manual steps to 3 reproducible deployment steps.

PROJECTS

- DubPatch (Voice AI System, Multilingual Dubbing, AI Pipeline):** Built an end-to-end pipeline for transforming English video/audio into multilingual dubbed outputs by orchestrating speech-to-text, translation, and neural TTS services. Automated a 4-stage manual workflow into a single reproducible pipeline, reducing manual intervention by 70%. Implemented asynchronous job handling and polling for external STT/TTS services, enabling processing of variable-length media inputs (30s-30+ min); modular ingestion/extraction/output (MP4/WAV) stages improved reusability and cut repeated setup effort by 3x.
Tech: Python, FastAPI, Async APIs, Audio Processing, Docker
- SignalFlow-rs (Rust DSP Engine, Audio Feature Extraction, Streaming Systems):** Built a library-first Rust audio DSP crate for end-to-end log-Mel feature extraction, with reusable modules for WAV/audio input, preprocessing, framing, windowing, spectrum computation, Mel projection, cached execution, streaming extraction, stress testing, examples, and Criterion benchmarks. Designed binary, examples, and benches to reuse the same `signalflow-rs` library API for better testability and benchmark consistency.
Tech: Rust, Cargo, DSP, FFT, Log-Mel Features, Streaming, Criterion
- GlyphPortrait (Generative Rendering, Computational Graphics, Image Processing):** Developed a typography-driven image rendering system that reconstructs portraits using procedurally arranged text glyphs. Implemented image preprocessing, contour extraction, semantic word placement, and rendering optimization workflows for stylized visual synthesis.
Tech: Python, OpenCV, NumPy, PIL, Matplotlib

OPEN SOURCE

- PyTorch AO PR #4332 and AI Infrastructure Contributions:** Contributed prototype bug fixes and code-quality improvements across PyTorch AO prototype modules, including RoPE fusion detection tests, quantization cleanup, embedding API/export cleanup, and int4 opaque tensor wording fixes. Also worked on benchmark diagnostics and reproducibility issues in AI systems evaluation workflows.
Tech: Python, PyTorch, Quantization, Benchmarking, Git, Linux

SKILLS SUMMARY

- Languages** : Rust, Python, C++, SQL
- ML/AI** : PyTorch, Transformers, DeepEval, LLM Evaluation, Scikit-learn
- Systems** : Rust DSP, FFT, Log-Mel Features, Streaming Pipelines, Benchmarking
- Infra** : Docker, Linux, REST APIs, Azure Container Registry, PostgreSQL, MySQL
- Tools** : Git, GitHub, Cargo, Jupyter
- Focus** : AI Inference, ML Systems, Agent Memory Systems, Computational Graphics

EDUCATION

- Anurag University** Hyderabad, India
Bachelor of Technology - Computer Science and Engineering; GPA: 3.26/4.00 July 2023 - Present
Courses: Operating Systems, Data Structures, Analysis of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases

HONORS AND AWARDS

- Awarded title of Networking Star by Founder of Codebasics - March, 2023

VOLUNTEER EXPERIENCE

- Community Lead at Microsoft AI Innovators Hub** Hyderabad, India
Conducted online and offline technical & soft-skills training impacting over 3000 students. Jan 2025 - August 2025
- Vice President at Malai AU Club** Hyderabad, India
Organized technical events and delivered workshops reaching over 500 students. July 2025 - April 2026